

REST API vs Graph API

Aspect	REST API	Graph API
Data Fetching	Multiple endpoints, each returns fixed data.	Single endpoint that allows flexible queries.
Request Structure	Uses multiple URLs/resources for different data.	Uses a single URL and queries specify data shape.
Over-fetching / Under-fetching	Often over-fetches or under-fetches data (fixed response structure).	Allows clients to specify exactly what fields and relationships to fetch, minimizing over/under-fetching.
Flexibility	Less flexible, adding/changing data requires new endpoints or versions.	Highly flexible, clients control queries dynamically.
Versioning	Often requires explicit API versioning.	Typically versioned less frequently, as schema evolves with queries.
Complexity	Simpler to implement and cache since fixed endpoints.	More complex query parsing and execution on server side.
Use cases	Best for simple CRUD operations with well-defined resources.	Best for complex data relationships and when clients need precise data shapes.
Example	<code>GET /users/123</code> returns user data.	<code>POST /graphql</code> with query <code>{ user(id:123) { name, posts { title } } }</code>

Summary

- **REST APIs** are resource-based, with multiple endpoints and fixed responses.
- **Graph APIs** provide a flexible query language allowing clients to specify exactly what data and nested relationships they want in a single request.